

A Scalable Distributed Pipeline for Wind Energy Forecasting

Alejandro Gómez, K. Ege Sural, Syed Ibrahim Hamza

1. Introduction and Motivation

The global transition toward sustainable energy systems is fundamentally dependent on our ability to harness weather patterns with high temporal and spatial precision. Unlike traditional fossil fuel-based power generation, wind energy is inherently variable, necessitating sophisticated forecasting to maintain electrical grid stability. The scaling of renewable energy reveals a critical challenge: without accurate supply forecasting, operators face escalating costs and heightened grid instability.

This project addresses the strategic challenge of processing over 600GB of heterogeneous, semi-structured observation streams to create an analysis-ready framework for energy forecasting. Wind energy modeling requires large-scale, high-frequency data, yet raw meteorological observations are often notoriously difficult to ingest at scale. Traditional data handling methods fail when confronted with the volume and complexity of global weather records, which often contain encoded strings and inconsistent reporting. By overcoming these data hurdles, we transform raw meteorological noise into structured, high-quality insights, directly impacting the accuracy of 24–72 hour renewable energy grid integration. To address this scale, we utilize the NOAA Integrated Surface Database (ISD) as our foundational ground truth.

2. Dataset Overview: NOAA Integrated Surface Database (ISD)

The NOAA Integrated Surface Database (ISD) serves as the primary repository for global meteorological research, providing hourly observations that constitute the "ground truth" for big data academic study. For this research, the ISD provides the high-resolution historical wind speed data required to model potential energy output over a multi-decadal horizon.

Specification	Details
Source	AWS Open Data (NOAA ISD)
Scale	600GB+ uncompressed (CSV format)
Coverage	35,000+ global stations; hourly observations
Temporal Scope	1995 – 2025
Geographic Scope	Contiguous United States (CONUS)
Primary Fields	WND, TMP, DEW, VIS, CIG, SLP

To ensure statistical rigor, we implemented a multi-stage filtering "funnel": starting with 28,474 global stations, we isolated 7,074 U.S. stations, then 6,225 CONUS stations, finally arriving at 4,943 valid stations with consistent reporting.

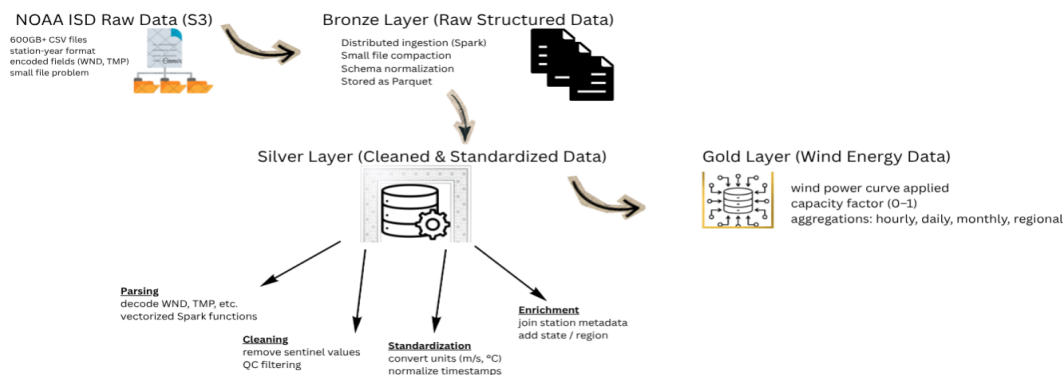
The "Encoded String Challenge" is most prominent in the WND field (e.g., "120,1,N,0045,1"). The pipeline uses vectorized Spark functions to split these strings, cast elements to numeric types, and filter by Quality Control (QC) flags. By applying these QC filters and excluding sentinel values, the pipeline achieves an 81% data retention rate. This 19% reduction represents the successful filtration of unreliable data, ensuring the Silver layer remains a trusted source of truth for energy modeling.

3. System Architecture and Medallion Data Lake Design

To maintain data quality and lineage, the pipeline utilizes a Medallion Architecture. This design pattern ensures that computationally intensive transformations are performed in a controlled, multi-stage environment.

3.1 The Medallion Layers

1. **Bronze Layer (Raw Structured):** This stage involves the distributed ingestion of raw CSV data from Amazon S3. We mitigate the small-file problem via compaction, merging thousands of fragmented files into optimized Parquet files. This ensures efficient partition pruning and schema normalization for downstream tasks.
2. **Silver Layer (Cleaned & Standardized):** The Silver layer performs the core engineering logic. It parses encoded strings into numeric features, handles sentinel values (e.g., 9999, 99999) by converting them to NULL, and standardizes measurements into SI units (m/s, $^{\circ}$ C, hPa).
3. **Gold Layer (Wind Energy Insights):** The final layer applies the theoretical wind power curve to standardized wind speeds, generating "Capacity Factors"—the primary metric for evaluating regional wind potential.



3.2 Technological Rationale

Distributed processing is mandatory for the 30-year window (1995–2025). While single-node tools like DuckDB were utilized for local development on specific subsets (CA, TX, MN, FL for 2018–2020), the full-scale processing of 800M+ rows necessitated PySpark on an AWS cluster (EC2 + S3). To maximize performance, the pipeline was implemented without Python UDFs (User Defined Functions), relying instead on native Spark functions to ensure "Spark-safe" performance across the cluster.

Tool	Function	Rationale
PySpark	Distributed Processing	Essential for processing 812M+ rows and 30 years of data across an AWS cluster; avoids Python UDF performance penalties.
DuckDB	Single-node Processing	Selected for local experimentation, rapid iteration on state-level subsets (CA, TX, MN, FL), and low-overhead benchmarking.
AWS S3 & EC2	Storage and Compute	Provides the scalable cloud infrastructure required for global weather records and multi-node Spark clusters.
Apache Airflow	Orchestration	Production-style scheduling layer used to manage task dependencies, traceability, and workflow observability.
UV	Environment Management	Manages isolated environments for Airflow tasks and dependencies, ensuring technical reproducibility across different runtimes.

The transition from high-level architecture to specific dataset engineering is driven by the extreme complexity of the primary source, particularly the millions of encoded WND strings that define the "small-file problem."

4. The Engineering Pipeline: Parsing and Data Enrichment

A project of this scale requires a config-driven pipeline to ensure reproducibility across cloud environments. By utilizing YAML-based configuration layers, we abstract S3 bucket names and EC2 hostnames, allowing the system to remain environment-agnostic.

The parsing of the WND field is a critical engineering task. Using vectorized Spark functions, the pipeline splits encoded strings by commas, casts the resulting elements to numeric types, and filters by Quality Control (QC) flags. Beyond wind, the pipeline successfully parsed and standardized TMP, DEW, VIS, CIG, and SLP.

During this ingestion phase, the pipeline achieved an 81% data retention rate. From an architectural perspective, this is framed as Data Integrity via QC-Aware Ingestion; the 19% loss represents the successful filtering of sentinel values and failed QC checks, ensuring the Silver layer remains a trusted source of truth. Finally, the data is enriched by joining station identifiers with metadata to provide geographic context, such as state and regional attributes.

5. Wind Power Curve and Capacity Factor

To convert meteorological observations into energy potential, we apply the physics of wind power using a piecewise power curve. The power output (P) as a function of wind speed (v) is defined by the following piecewise function:

$$P(v) = \begin{cases} 0 & v < v_{cut-in} \\ \left(\frac{v - v_{cut-in}}{v_{rated} - v_{cut-in}} \right)^3 & v_{cut-in} \leq v < v_{rated} \\ 1 & v_{rated} \leq v \leq v_{cut-out} \\ 0 & v > v_{cut-out} \end{cases}$$

Our model adopts the following physical constants and generic utility-scale turbine assumptions:

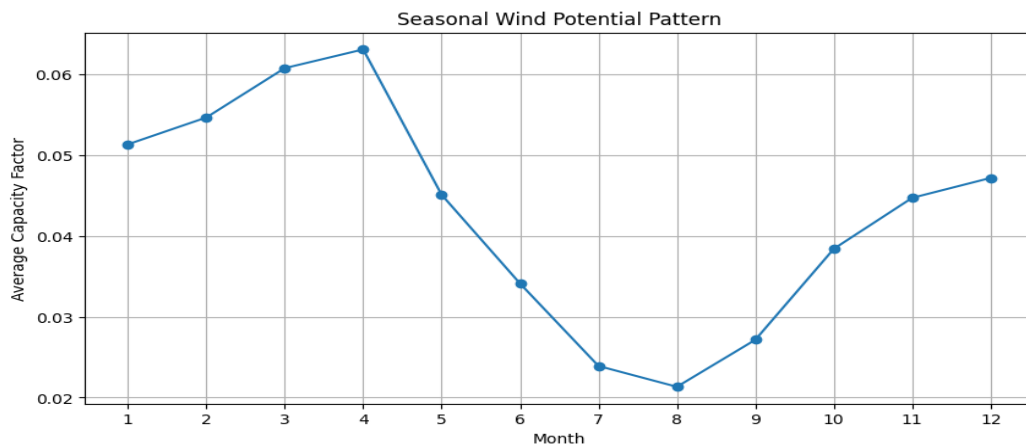
- Air Density: 1.225 kg/m^3
- Cut-in Speed (V_{cut}): 3.5 m/s
- Rated Speed (V_{rated}): 13.0 m/s
- Cut-out Speed (V_{cut_out}): 25.0 m/s

The resulting Capacity Factor (ranging from 0 to 1) provides a normalized metric for comparing wind potential across diverse geographies, independent of specific turbine configurations.

6. Analytical Results: Temporal and Spatial Wind Trends

The Gold Layer transformations allow for high-level strategic insights into the wind potential of the contiguous United States.

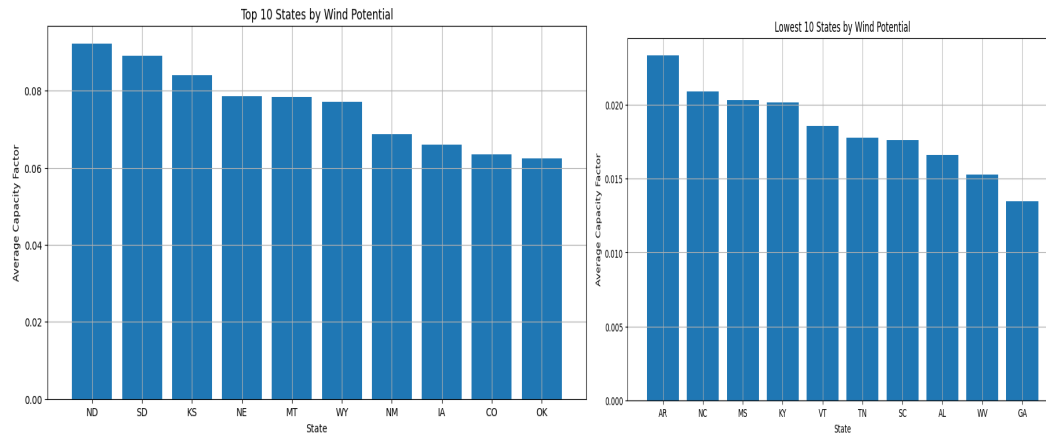
6.1 Seasonal Patterns



Analysis of seasonal trends (Q1) shows that the average capacity factor peaks in April (~ 0.06) and reaches its low in August (~ 0.02). This summer trough creates a "strategic energy gap" in the U.S.,

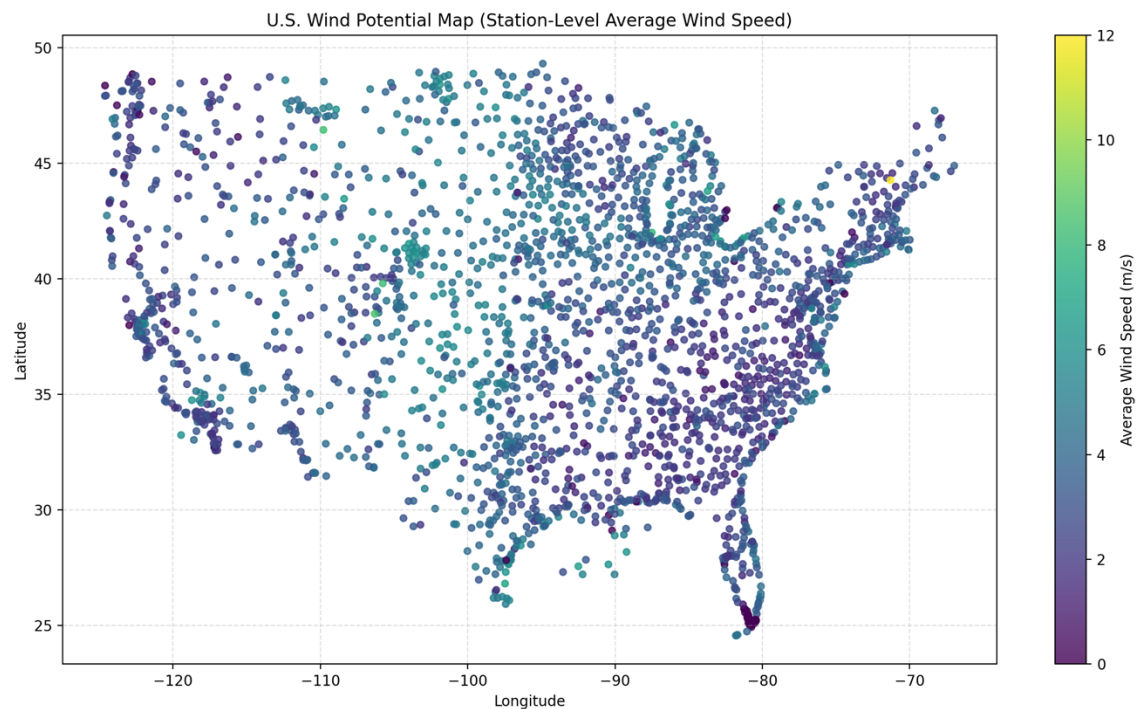
as the decline in wind potential coincides with peak cooling demand, highlighting the critical need for solar balancing or long-duration storage.

6.2 Spatial Patterns

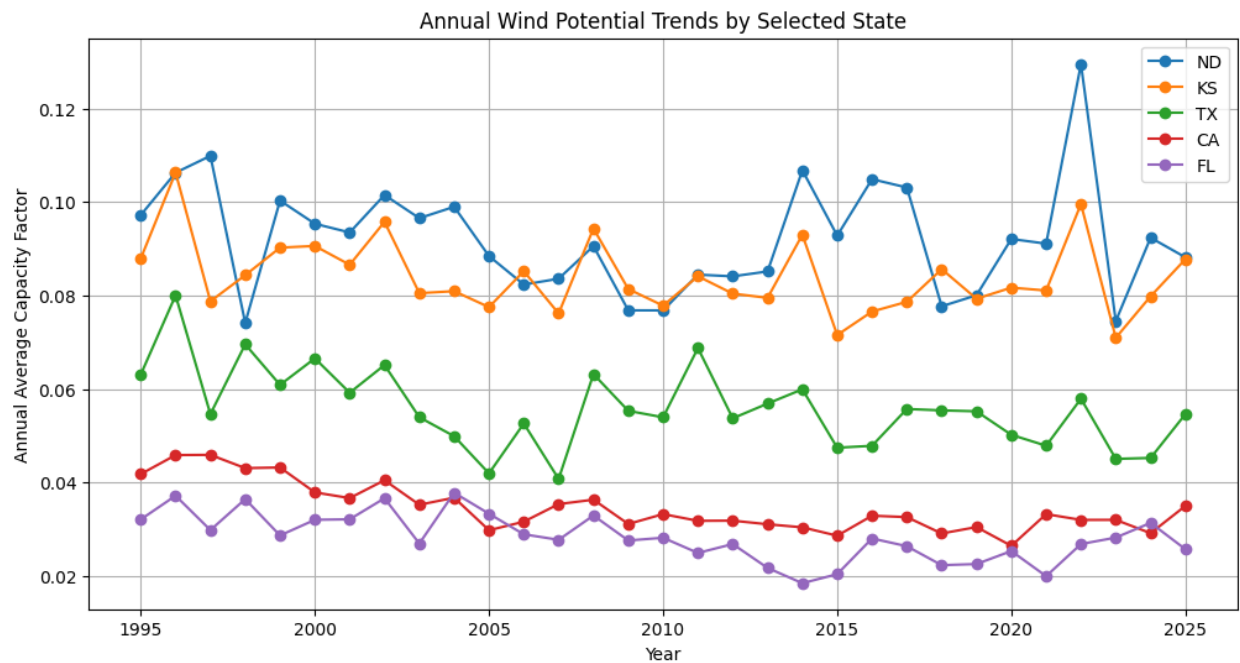


Spatial analysis (Q2) confirms the geographic dominance of the Great Plains. The top five states by wind potential are North Dakota (ND), South Dakota (SD), Kansas (KS), Nebraska (NE), and Montana (MT). Conversely, the states with the lowest potential are Georgia (GA), West Virginia (WV), Alabama (AL), South Carolina (SC), and Tennessee (TN)—must rely on alternative energy mixes due to their geographic limitations.

6.3 U.S. Wind Potential Map

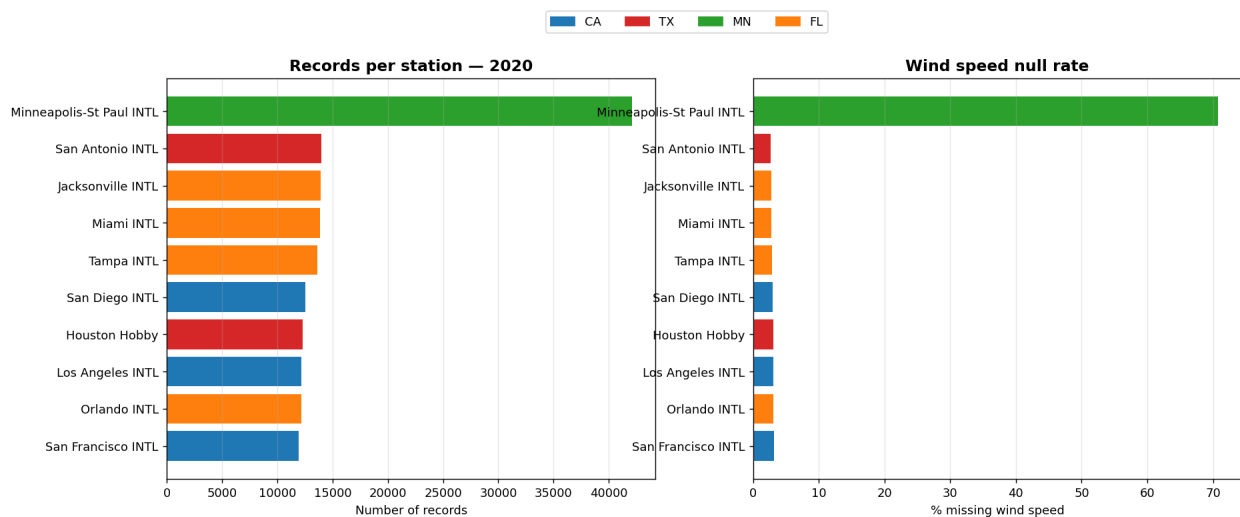


6.4 Long-term Trends



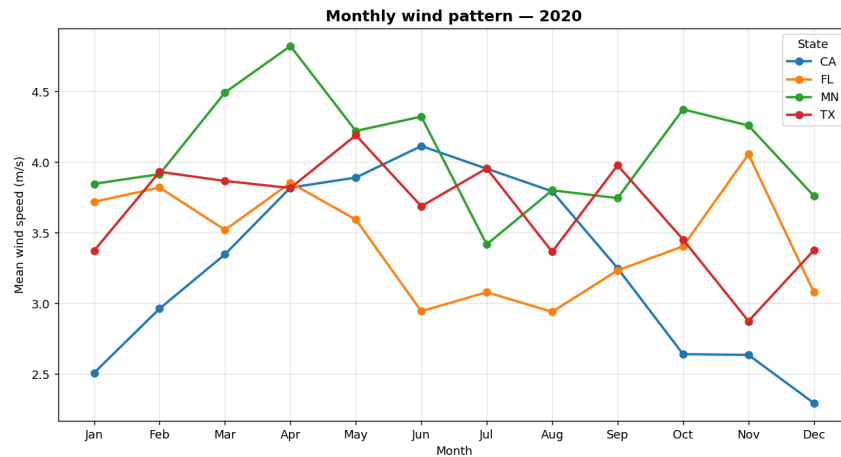
Annual wind potential (Q3) from 1995 to 2025 remains remarkably stable in high-potential states like ND, KS, and TX. This long-term consistency underscores the viability of wind as a foundational component of long-term infrastructure investment.

6.5 Data Volume and Quality Analysis of US Weather Stations (2020)



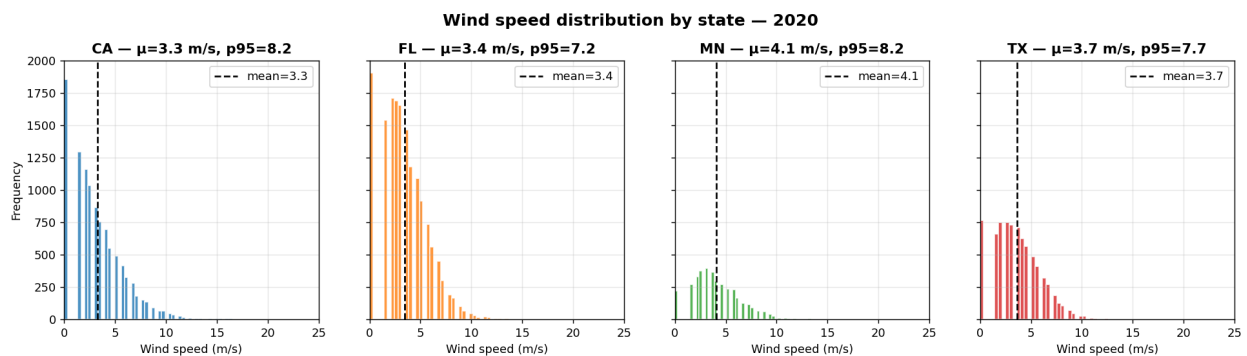
These charts compare the total number of data records against the percentage of missing wind speed information across various US airport stations in 2020. While Minneapolis-St Paul INTL shows the highest volume of records, it also exhibits a disproportionately high null rate for wind speed data compared to the other locations.

6.6 Monthly wind pattern



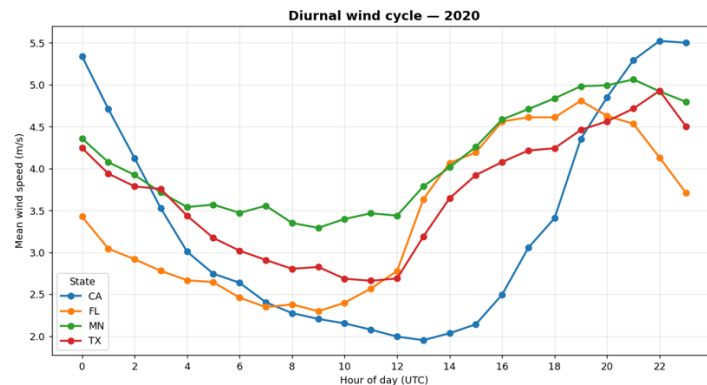
This line graph illustrates the monthly fluctuations in mean wind speeds across California, Florida, Minnesota, and Texas throughout the year 2020. The data reveals that while Minnesota generally maintained the highest wind speeds, all four states exhibited distinct seasonal trends with significant variations in peak intensity.

6.7 Wind Speed Distribution by State (2020)



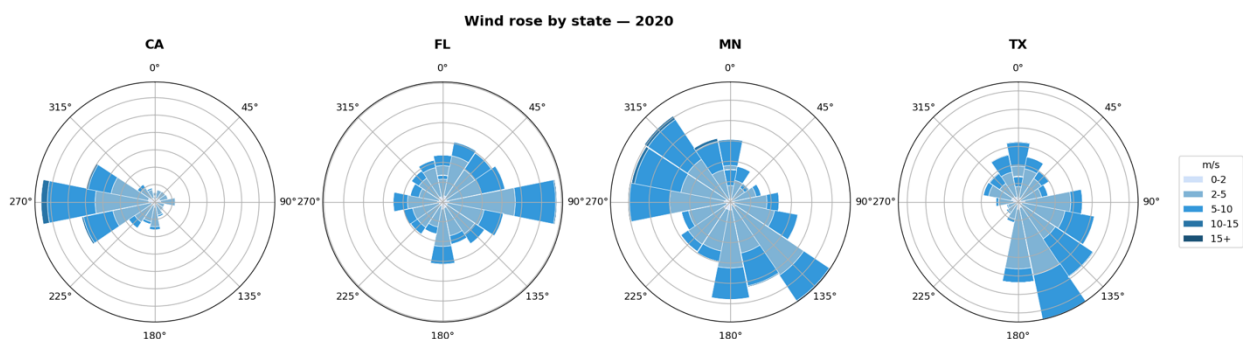
These histograms illustrate the statistical spread of wind speed occurrences across the four analyzed states. The data shows that while wind speeds are generally right-skewed, Minnesota (MN) exhibits the highest mean velocity and a significantly broader distribution of high-speed events compared to the more concentrated, lower-speed profiles of California and Florida.

6.8 Diurnal Wind Cycle (2020)



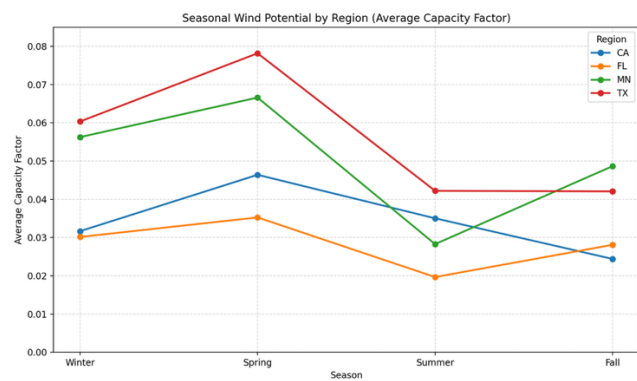
This line graph visualizes the average hourly fluctuations in wind speed over a 24-hour UTC period, identifying critical daily patterns for grid stability. The results reveal a consistent "u-shaped" diurnal cycle across all states, where wind intensity typically reaches its daily minimum during the midday hours (12:00–14:00 UTC) and peaks during the late night and early morning.

6.9 Wind Directional Analysis (Wind Roses)



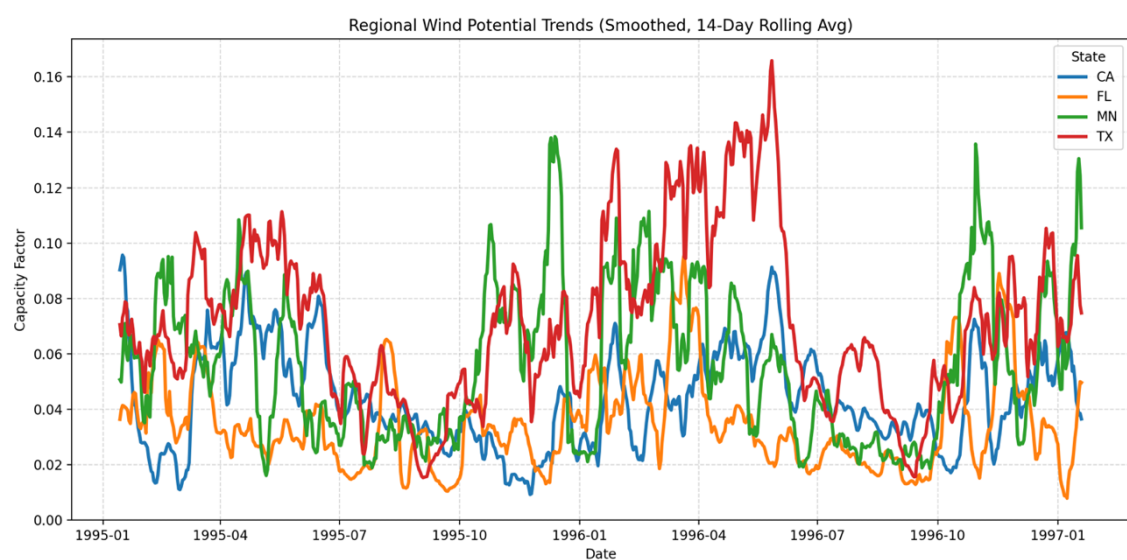
These circular "wind rose" plots map the frequency of wind direction and speed intensity, providing essential spatial context for turbine orientation. The analysis highlights distinct regional patterns, such as the dominant westerly winds in California (CA) and the strong southerly-to-southeasterly flow characterizing the wind profile in Texas (TX).

6.10 Seasonal Wind Potential by Region (Average Capacity Factor)



Texas (TX) consistently maintains the highest seasonal capacity factor, while Minnesota (MN) shows the most significant seasonal variance, peaking sharply in Spring before dropping to its minimum in Summer.

6.11 Regional Wind Potential Trends (14-Day Rolling Average)

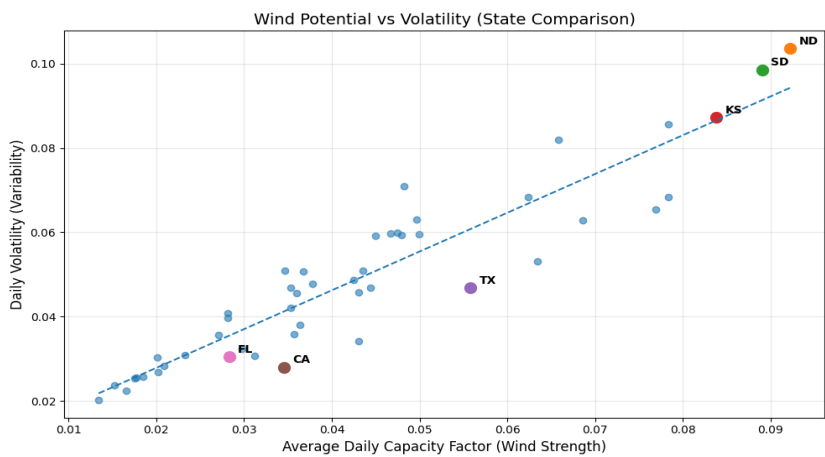


This time-series visualization displays the 14-day rolling average of capacity factors for California, Florida, Minnesota, and Texas from early 1995 through 1997. The smoothed data highlights regional fluctuations and seasonal variations in wind energy potential, providing the historical depth necessary for evaluating supply stability across diverse geographies.

7. Advanced Analytics: Volatility, Stability, and Extreme Events

Average potential is an incomplete metric without understanding volatility.

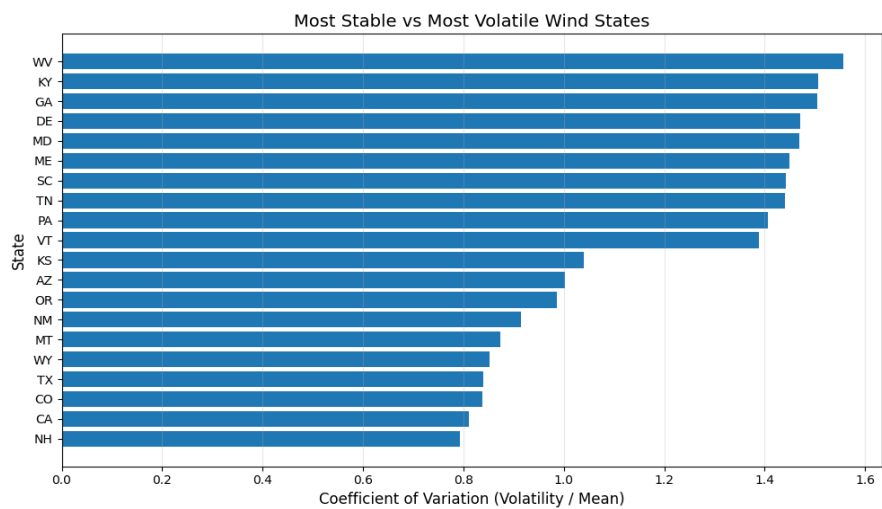
7.1 Volatility vs. Strength



The relationship between strength (Mean Capacity Factor) and variability (Daily Volatility) generally follows a linear trend. However, Texas (TX) emerges as a key strategic outlier; its volatility sits significantly below the national trend line relative to its strength, making it an ideal candidate for reliable grid integration.

7.2 Stability and the Coefficient of Variation

Using the Coefficient of Variation ($CV = \text{Volatility} / \text{Mean}$), we rank states by stability:



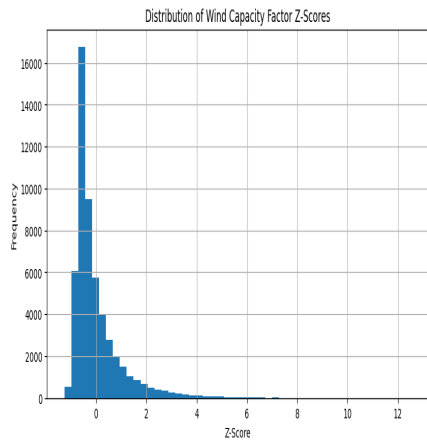
- Most Stable States: New Hampshire (NH), California (CA), and Colorado (CO).
- Most Volatile States: West Virginia (WV), Kentucky (KY), and Georgia (GA).

7.3 Extreme Events

We identify anomalies using a Z-score methodology, which normalizes daily fluctuations against state-level historical averages:

$$Z = (\text{daily capacity factor} - \text{state average}) / \text{state standard deviation}$$

While the distribution confirms that the vast majority of days are "normal," the pipeline effectively identifies "high_wind" and "low_wind" Z-score outliers. This categorization is essential for preparing the grid for potential surges or supply deficits.



8. Predictive Analytics: Machine Learning Framework

The pipeline transitions from retrospective analysis to proactive forecasting by focusing on 24–72 hour windows, allowing grid operators to manage supply fluctuations effectively.

Feature Engineering and Temporal Splits

The predictive framework utilizes an ML table built from the Gold layer. We utilized a strict time-based split to ensure model validity: Train (≤ 2019), Validation (2020–2022), and Test (≥ 2023). Feature engineering includes:

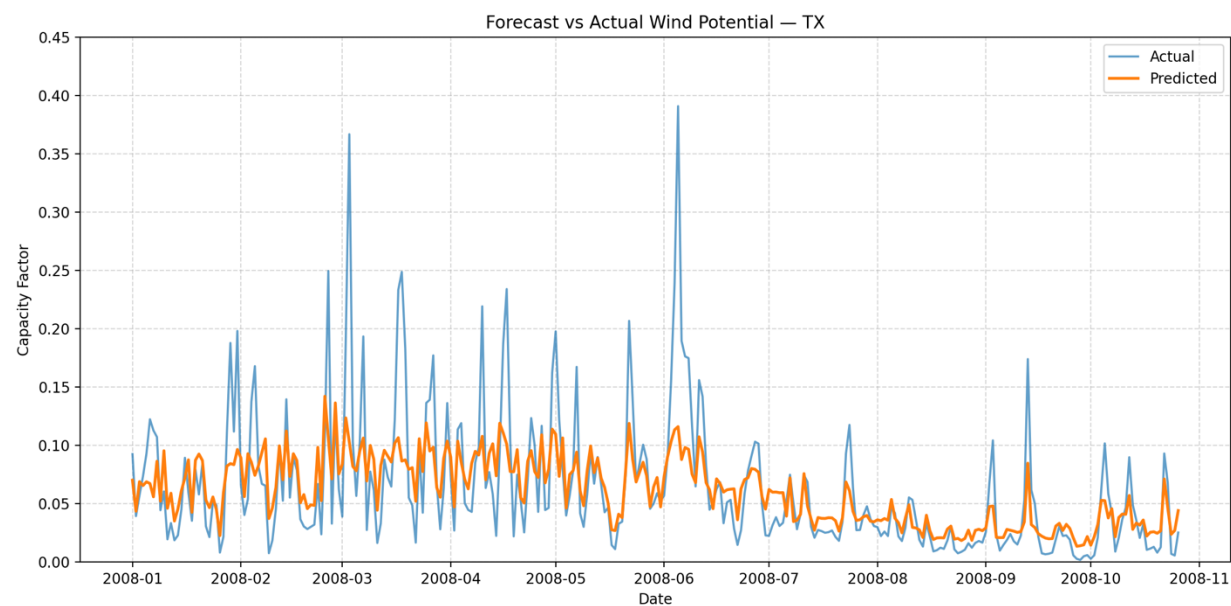
- Lag features and rolling statistics to capture temporal trends.
- Weather Aggregates: TMP (Temperature), DEW (Dew Point), VIS (Visibility), CIG (Ceiling), and SLP (Sea Level Pressure).

Model Performance Comparison

Model	Status	RMSE	MAE
Baseline	Reference	0.065	0.048
Linear Regression	Evaluated	0.051	0.034
Random Forest	Evaluated	0.046	0.029
Gradient Boosted Trees (GBT)	Best Model	0.042	0.025

The GBT model demonstrated the highest precision for forecasting supply potential. These models are managed via an automated registry for production deployment.

8.1 Model Validation: Forecast vs. Actual Wind Potential (TX)



The graph demonstrates the Gradient Boosted Trees (GBT) model's capability to track real-world wind fluctuations, providing a practical benchmark for its accuracy in 24–72 hour supply forecasting.

8.2 Detailed Model Evaluation and Feature Attribution

To ensure the reliability of the 24–72 hour forecasts, the Gradient Boosted Trees (GBT) model was subjected to a rigorous evaluation within the experimental notebook environment. This phase focuses on quantifying error metrics and identifying the most influential meteorological drivers.

8.2.1 Feature Importance and Temporal Drivers

The model’s predictive power is heavily influenced by temporal dependencies. As illustrated in the feature importance analysis, Lag features (representing past wind speeds) are the most critical predictors, outperforming raw weather aggregates.

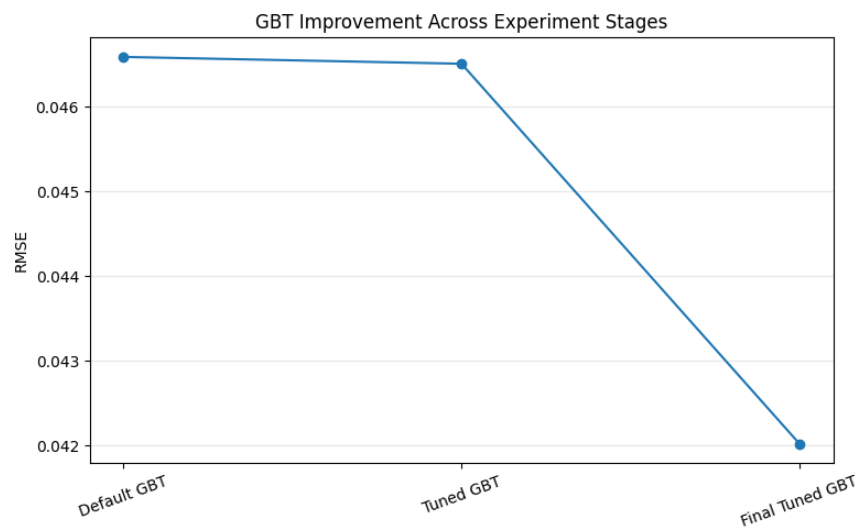


Figure 8.2: Feature Importance ranking, highlighting the dominance of lag variables and temperature (TMP) in the forecasting logic.

Beyond temporal lags, weather aggregates like Temperature (TMP) and Sea Level Pressure (SLP) provide essential physical context for shifting weather patterns.

8.2.2 Comparative Model Performance

Comparing the GBT model against baseline and linear benchmarks confirms its superior ability to handle non-linear meteorological relationships.

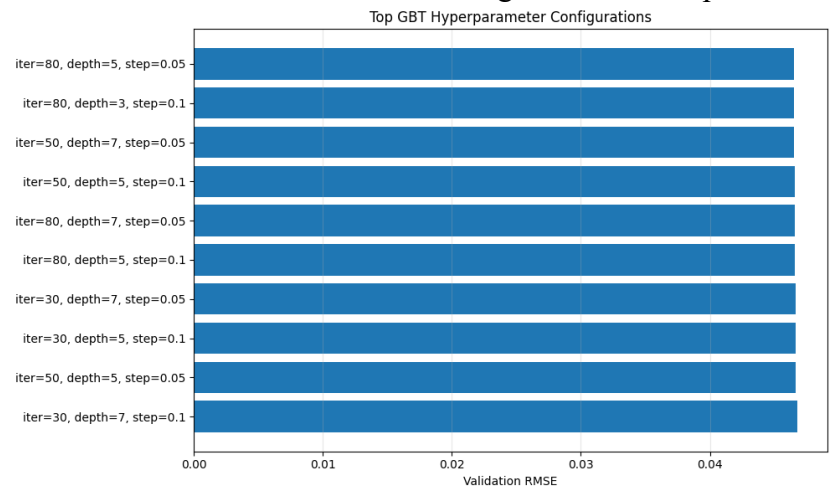


Figure 8.3: Comparative RMSE and MAE across models, validating GBT as the best-performing architecture.

The GBT model achieved a Root Mean Squared Error (RMSE) of 0.042 and a Mean Absolute Error (MAE) of 0.025, representing a significant improvement over the baseline reference.

8.2.3 Visual Validation of Forecast Accuracy

The model’s performance is further validated by its ability to track real-world fluctuations. Scatter and time-series analyses demonstrate a strong correlation between predicted and actual capacity factors.

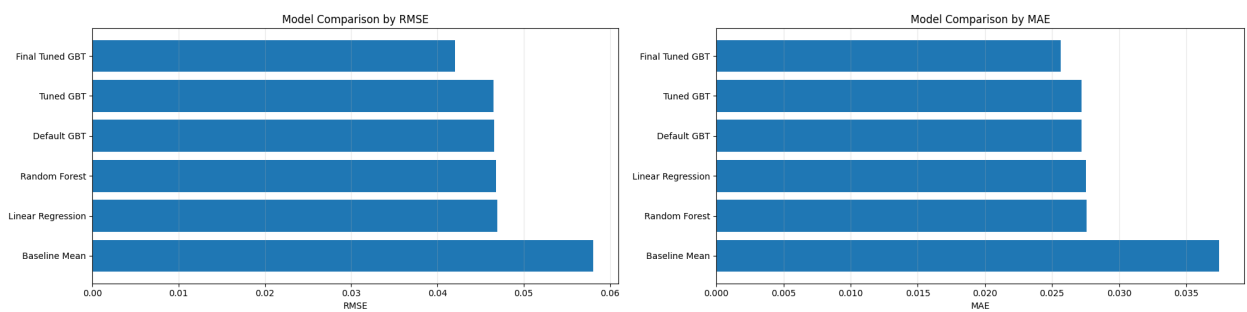
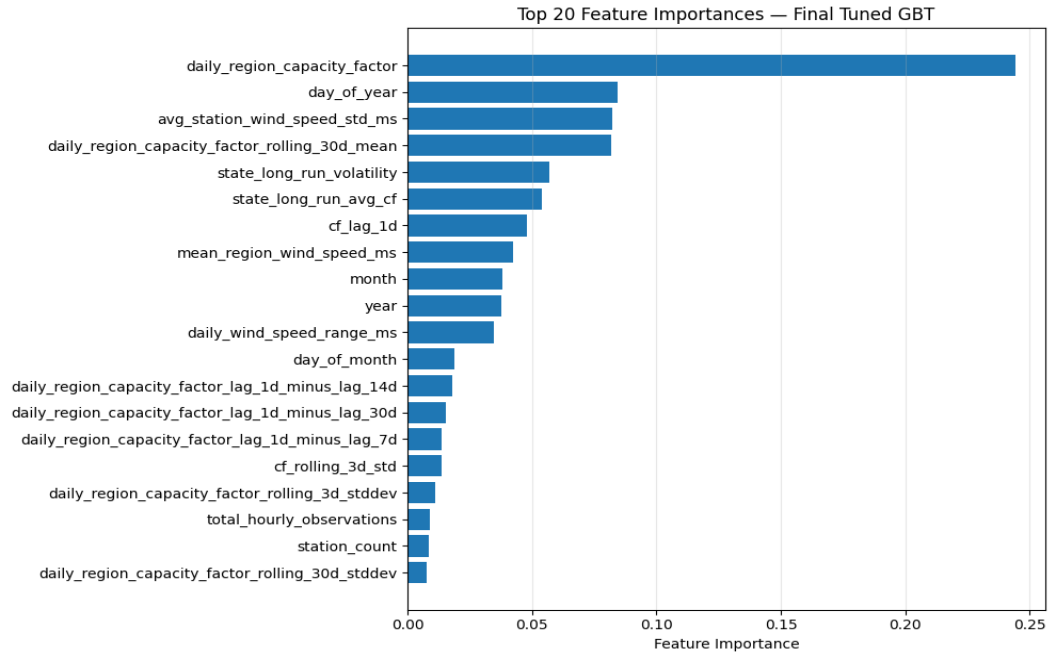


Figure 8.4: Scatter and Time-Series plots comparing Predicted vs. Actual wind potential, showcasing the model's precision in tracking seasonal and daily surges.

8.2.4 Strategic Insight: Volatility vs. Strength

Integrating machine learning insights with state-level analysis reveals that the relationship between wind strength and variability remains largely linear across the U.S..



8.5: Wind Potential vs. Volatility by state, confirming Texas (TX) as a strategic outlier with high strength and disproportionately low volatility.

9. Production Orchestration and Benchmarking

To move beyond manual execution, the pipeline utilizes Apache Airflow to simulate a production-grade workflow. This orchestration layer provides critical observability through the Airflow UI, allowing for real-time monitoring of task dependencies and system health.

9.1 Pipeline Orchestration (wind_pipeline_dag)

The workflow is managed through the wind_pipeline_dag, which coordinates the data's journey from raw ingestion to final validation.

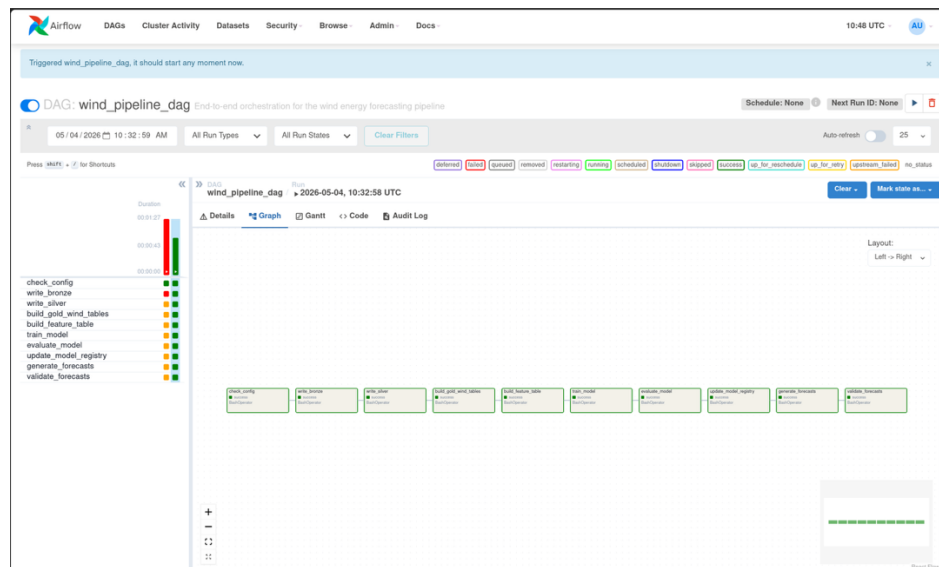


Figure 9.1: The linear DAG structure showing the successful execution of all ten pipeline stages.

The sequence consists of ten key tasks:

- `check_config`: Verifies environment variables and configuration layers.
- `write_bronze`: Orchestrates raw data ingestion from Amazon S3.
- `write_silver`: Performs vectorized parsing and QC filtering.
- `build_gold_wind_tables`: Calculates physical Capacity Factors.
- `build_feature_table`: Constructs the ML-ready feature set.
- `train_model` & `evaluate_model`: Trains and assesses the GBT forecasting models.
- `update_model_registry`: Automates model deployment.
- `generate_forecasts` & `validate_forecasts`: Produces and benchmarks 24–72 hour supply predictions.

9.2 Observability and Execution Metrics

The Airflow UI serves as a "control tower" for the pipeline, providing detailed metrics on run reliability and task duration. As shown in the DAG Runs Summary, the system tracks success and failure rates to ensure high availability.

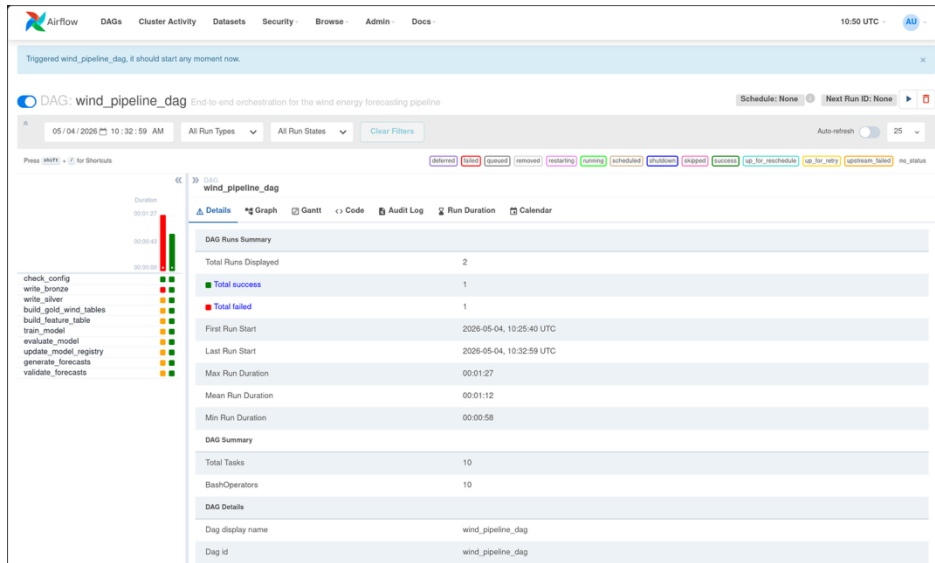


Figure 9.1: DAG Run metrics indicating successful execution history and task counts.

To maintain a reproducible environment, the pipeline utilizes YAML-based configuration abstracted from the core logic. The `check_config` task validates these external parameters (e.g., identifying the active user config as `configs/users/syed.yaml`) before processing begins.

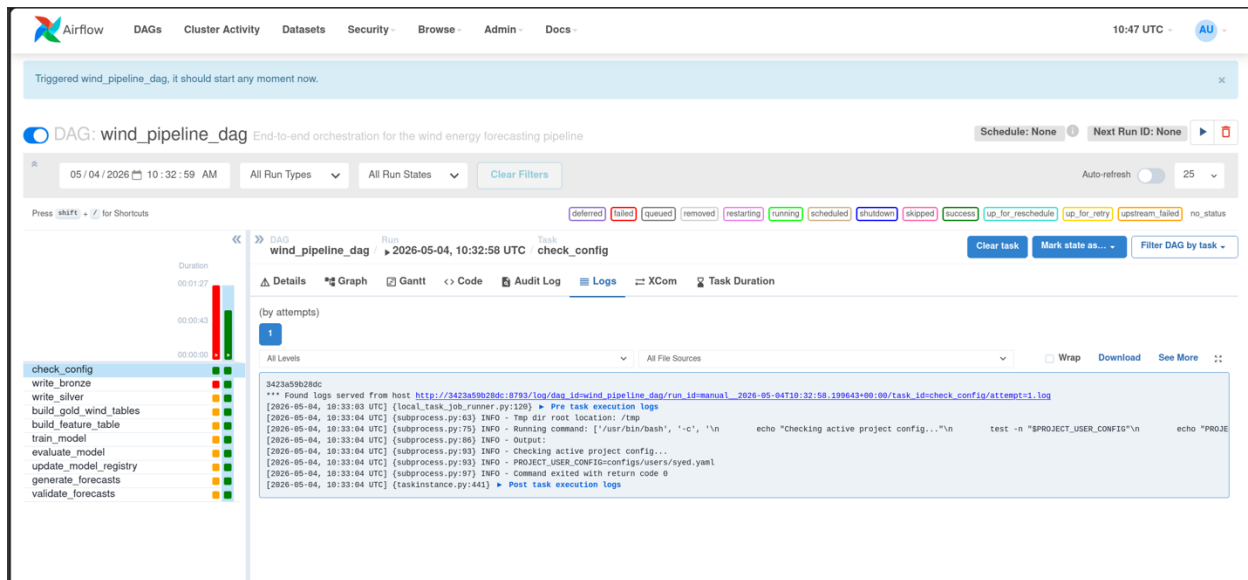


Figure 9.2: Execution logs for the `check_config` task verifying active project configurations.

Furthermore, the system supports "dry run" logic for safety, allowing developers to skip intensive ingestion layers during testing phase without breaking the DAG's logical flow.

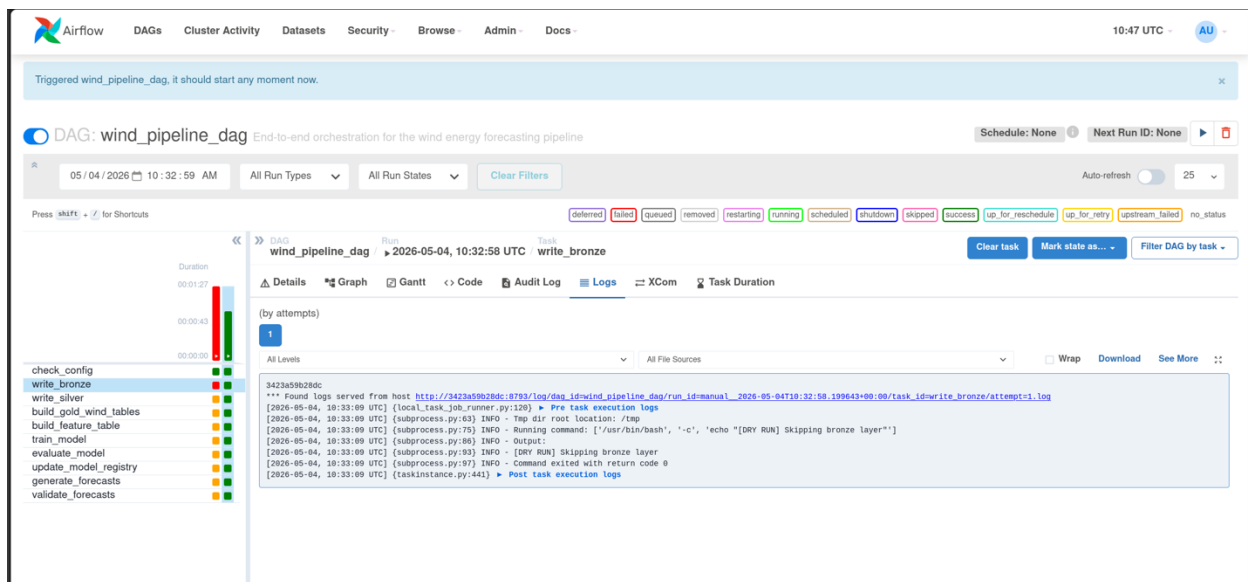


Figure 9.4: Task logs demonstrating the [DRY RUN] skip logic for the write_bronze layer.

Finally, the Gantt View provides a temporal breakdown of the pipeline, identifying that tasks like write_silver and train_model represent the most computationally intensive stages, whereas configuration checks are near-instantaneous.

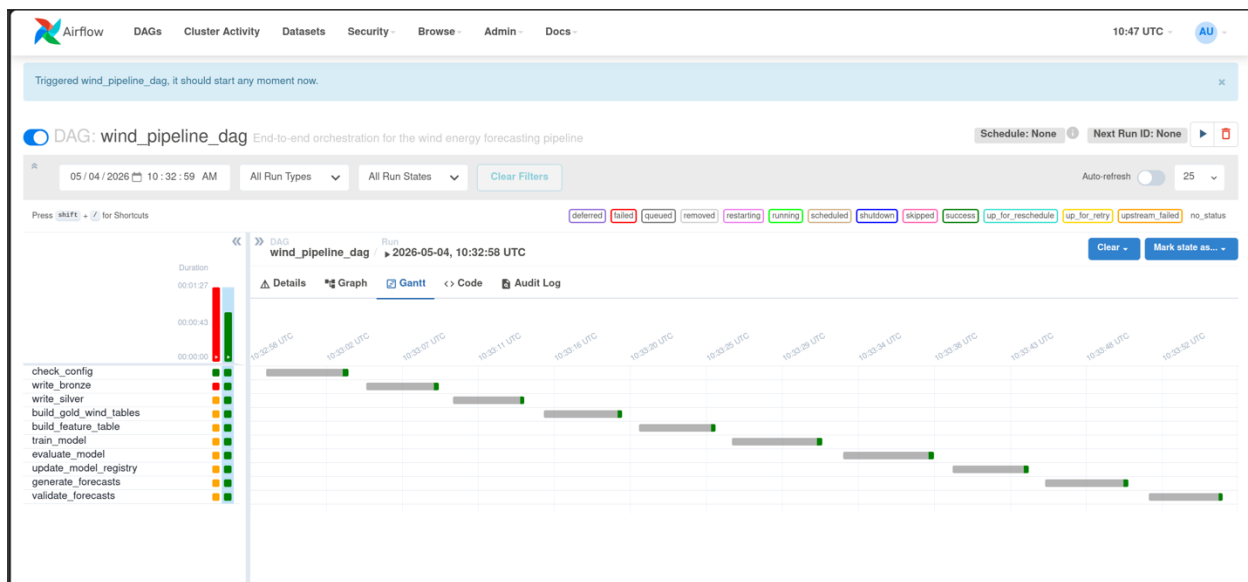


Figure 9.5: Gantt chart visualizing the relative execution time for each task in the pipeline.

9.3 Comparative Performance Analysis

To quantify the "near-zero startup cost" of the local development engine, we conducted a series of benchmarks across three core tasks: filter_year_region, daily_region_aggregation, and grouped_temporal_summaries.

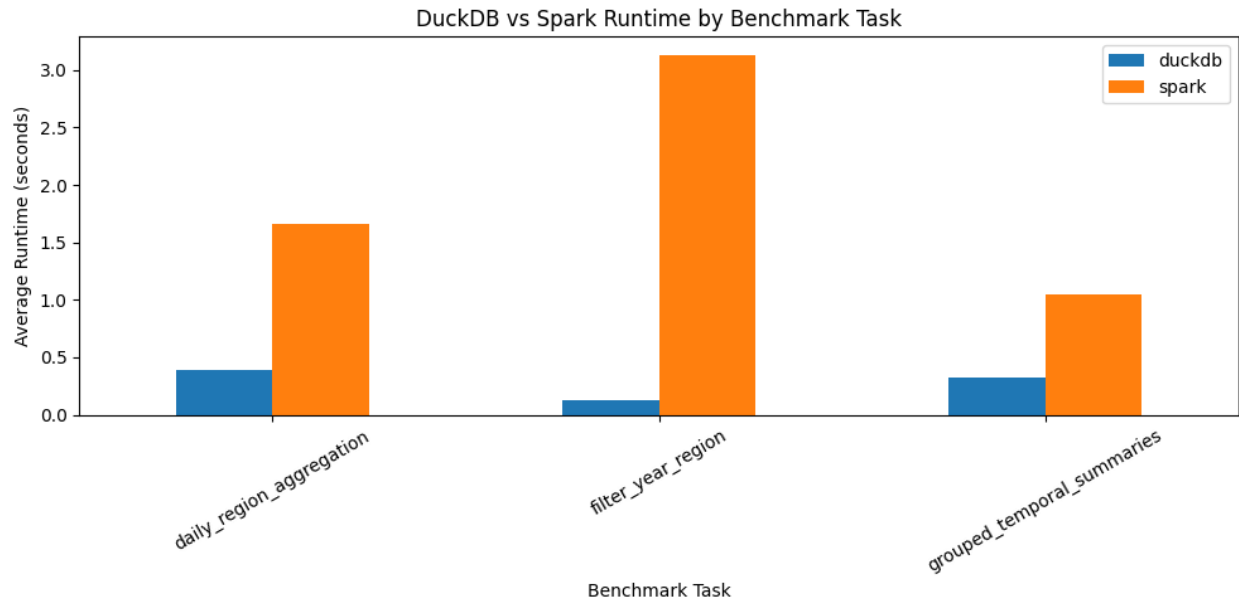


Figure 9.6: Average Runtime (seconds) by Benchmark Task.

As illustrated in Figure 9.6, DuckDB consistently maintains an average runtime of under 0.5 seconds for all local operations. In contrast, Spark’s average runtime ranges from 1.0 to over 3.0 seconds, even when processing identical state-level subsets.

9.3.1 Overhead and Runtime Ratios

The performance gap is most pronounced in the `filter_year_region` task, which involves initial data access and basic filtering.

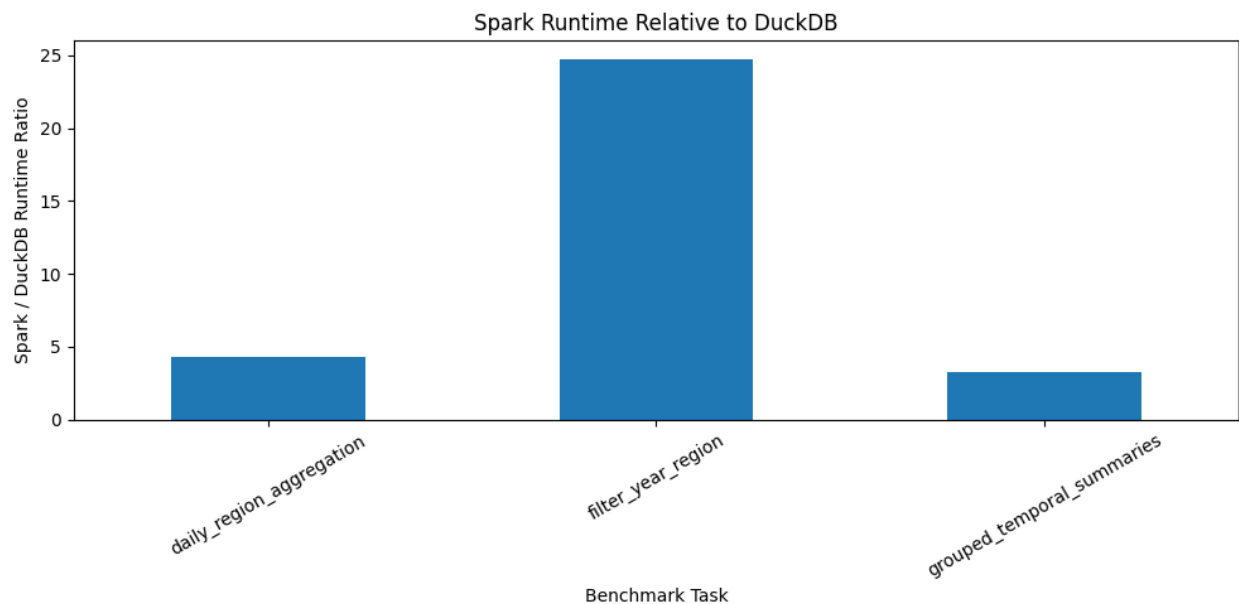


Figure 9.7: Spark Runtime Relative to DuckDB (Runtime Ratio).

The analysis of the Spark / DuckDB Runtime Ratio (Figure 9.7) reveals that Spark is approximately 25 times slower than DuckDB for simple filtering tasks on local data. This discrepancy is attributed to the "startup cost" of the Spark JVM and its distributed execution planner, which introduces significant latency compared to DuckDB's in-process, vectorized C++ engine.

9.3.2 Statistical Variance and Stability

The Runtime Distribution further highlights the stability of the local engine versus the distributed framework.

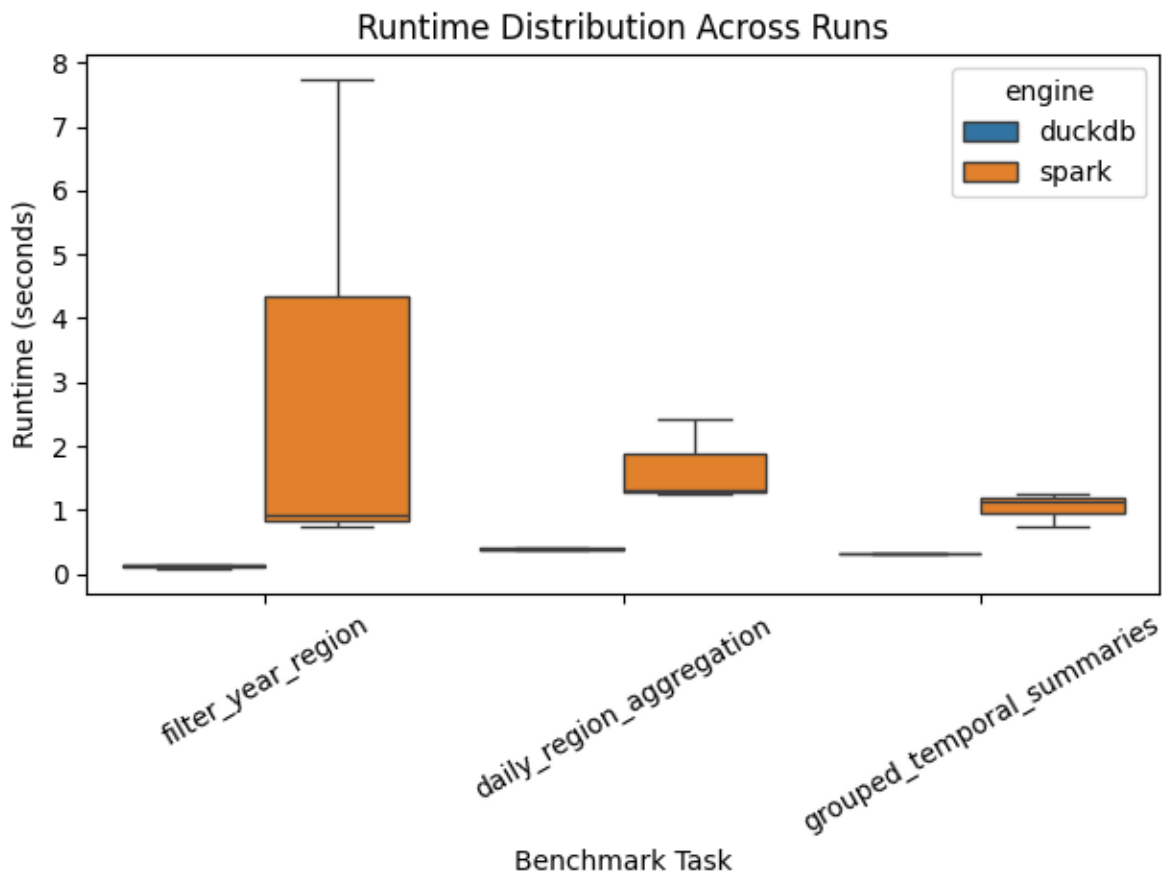


Figure 9.8: Box plot of Runtime Distribution Across Multiple Runs.

- DuckDB: Shows almost zero variance (visible as thin lines in Figure 9.8), making it ideal for rapid, iterative experimentation.
- Spark: Exhibits high variability and significant outliers. In the filter_year_region task, Spark's execution time fluctuated between 0.8 and nearly 8.0 seconds.

While these local benchmarks highlight DuckDB's superiority for rapid iteration on subsets (e.g., CA, TX), Spark remains the mandatory choice for the 812M+ row scale, as single-node systems hit memory limits that Spark's distributed architecture on AWS successfully overcomes.

10. Conclusion and Final Achievements

This project successfully navigated the complex journey from 600GB of heterogeneous, semi-structured observation streams to a structured, insight-driven Gold dataset. By bridging the gap between raw meteorological data, physical energy modeling, and distributed system engineering, we have established a robust foundation for renewable energy planning and grid stability.

Final Achievements:

- **End-to-End Scalability via Medallion Architecture:** Built a multi-stage pipeline (Bronze, Silver, Gold) capable of processing 30 years of global data. The system successfully managed 812,991,212 hourly station rows and 19,430,672 daily rows, ensuring data integrity through QC-aware ingestion.
- **Operational Excellence with Apache Airflow:** Transitioned from manual execution to a production-grade environment using Airflow. The `wind_pipeline_dag` provides full observability across ten key tasks—from configuration checks to forecast validation—ensuring technical reproducibility and system health monitoring.
- **Performance Engineering & Hybrid Processing:** Quantified the architectural trade-offs between local and cloud engines. Benchmarking proved that DuckDB offers a 25x speed advantage for local prototyping and rapid iteration, while Apache Spark remains the mandatory backbone for processing multi-decadal, 800M+ row datasets on AWS.
- **High-Precision Predictive Forecasting:** Developed and validated a Gradient Boosted Trees (GBT) model specifically tuned for 24–72 hour windows. The model achieved a Root Mean Squared Error (RMSE) of 0.042, significantly outperforming linear and baseline references by leveraging temporal lag features and weather aggregates.
- **Strategic Energy Insights:** Spatial and temporal analysis identified the Great Plains (ND, SD, KS) as the primary hub for U.S. wind potential. Crucially, Texas (TX) was identified as a strategic outlier, exhibiting high wind strength with disproportionately low volatility, making it an ideal candidate for reliable long-term grid integration.

The resulting Gold tables and predictive frameworks are optimized for production deployment, providing the historical depth and real-time accuracy necessary for the global transition toward sustainable energy systems.